

1. Laplacian Filter

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
image = cv2.imread("img_3.jpg", cv2.IMREAD_GRAYSCALE)
if image is None:
    print("Error: Could not read image.")
    exit()
kernel1 = np.array([[0, -1, 0], [-1, 5, -1], [0, -1, 0]], dtype=np.float32)
kernel2 = np.array([[-1, -1, -1], [-1, 9, -1], [-1, -1, -1]], dtype=np.float32)
output1 = cv2.filter2D(image, -1, kernel1)
output2 = cv2.filter2D(image, -1, kernel2)
plt.figure(figsize=(10, 8))
titles = [
    "Original Image", "Kernel 1", "Kernel 2",
]
images = [image, output1, output2]
for i in range(3):
    plt.subplot(2, 2, i + 1)
    plt.imshow(images[i], cmap="gray")
    plt.title(titles[i])
    plt.axis("off")
plt.tight_layout()
plt.show()
```

OUTPUT

Original Image



Kernel 1



Kernel 2



2. Unsharp Masking

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
image = cv2.imread("img_3.jpg", cv2.IMREAD_GRAYSCALE)
if image is None:
    print("Error: Could not read image.")
    exit()
blur = cv2.GaussianBlur(image, (9, 9), 2)
mask = cv2.subtract(image, blur)
sharpened = cv2.addWeighted(image, 1.0, mask, 1.0, 0)
plt.figure(figsize=(10, 8))
titles = ["Original Image", "Blurred Image", "Unsharp Mask", "Sharpened Image"]
images = [image, blur, mask, sharpened]
for i in range(4):
    plt.subplot(2, 2, i + 1)
    plt.imshow(images[i], cmap="gray")
    plt.title(titles[i])
    plt.axis("off")
plt.tight_layout()
plt.show()
```

OUTPUT

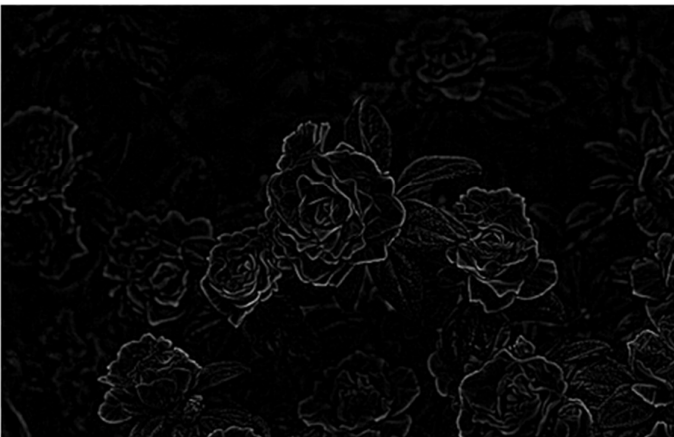
Original Image



Blurred Image



Unsharp Mask



Sharpened Image



3. High Boosting

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
image = cv2.imread("img_3.jpg", cv2.IMREAD_GRAYSCALE)
if image is None:
    print("Error: Could not read image.")
    exit()
blur = cv2.GaussianBlur(image, (9, 9), 2)
mask = cv2.subtract(image, blur)
A_values = [1, 1.5, 2, 3]
results = []
for A in A_values:
    high_boost = cv2.addWeighted(image, A, mask, 1, 0)
    results.append(high_boost)
plt.figure(figsize=(12, 8))
plt.subplot(2, 3, 1)
plt.imshow(image, cmap="gray")
plt.title("Original Image")
plt.axis("off")
plt.subplot(2, 3, 2)
plt.imshow(blur, cmap="gray")
plt.title("Blurred Image")
plt.axis("off")
for i in range(4):
    plt.subplot(2, 3, i + 3)
    plt.imshow(results[i], cmap="gray")
    plt.title(f"High Boost (A={A_values[i]})")
    plt.axis("off")
plt.tight_layout()
plt.show()
```

OUTPUT

Original Image



Blurred Image



High Boost (A=1)



High Boost (A=1.5)



High Boost (A=2)



High Boost (A=3)



4. High Boost Filtering Using Laplacian Operator

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
image = cv2.imread("img_1.jpg", cv2.IMREAD_GRAYSCALE)
if image is None:
    print("Error: Could not read image.")
    exit()
A_values = [1, 2, 3, 5]
plt.figure(figsize=(14, 10))
plt.subplot(3, 3, 1)
plt.imshow(image, cmap="gray")
plt.title("Original Image")
plt.axis("off")
for i, A in enumerate(A_values):
    kernel1 = np.array([[0, -1, 0], [-1, A + 4, -1], [0, -1, 0]],
dtype=np.float32)
    output = cv2.filter2D(image, -1, kernel1)
    plt.subplot(3, 3, i + 2)
    plt.imshow(output, cmap="gray")
    plt.title(f"4-Nbr A={A}")
    plt.axis("off")
for i, A in enumerate(A_values):
    kernel2 = np.array([[-1, -1, -1], [-1, A + 8, -1], [-1, -1, -1]],
dtype=np.float32)
    output = cv2.filter2D(image, -1, kernel2)
    plt.subplot(3, 3, i + 6)
    plt.imshow(output, cmap="gray")
    plt.title(f"8-Nbr A={A}")
    plt.axis("off")
plt.tight_layout()
plt.show()
```

OUTPUT

